

5. The photograph shows a koala (*Phascolarctos cinereus*). These animals live in trees in Australia. Their chromosome number is 16.



(a) (i) **Complete Table 5** about cell division in the koala. [3]

Table 5

	Number
Number of chromosomes in one body cell.	16
Number of cells produced from one cell by mitosis .	
Number of chromosomes in each new cell produced by mitosis .	
Number of chromosomes in each new cell produced by meiosis .	

(ii) State:

I. **one** function of mitosis; [1]

.....

II. the scientific name for the daughter cells produced by meiosis. [1]

.....



- (b) Another Australian animal, the grey kangaroo (*Macropus giganteus*) also has 16 chromosomes but its appearance is very different from the koala.



Underline which **one** of the following statements explains what causes the appearance of these two species to be different. [1]

- They live in different habitats.
- The genes on their chromosomes are different.
- Their diet consists of different types of plants.

3400U201
11

6

